

ORF309/MAT380

Poisson Processes III: Examples

(pp. 106-110)

Ioannis Akrotirianakis

Agenda for today

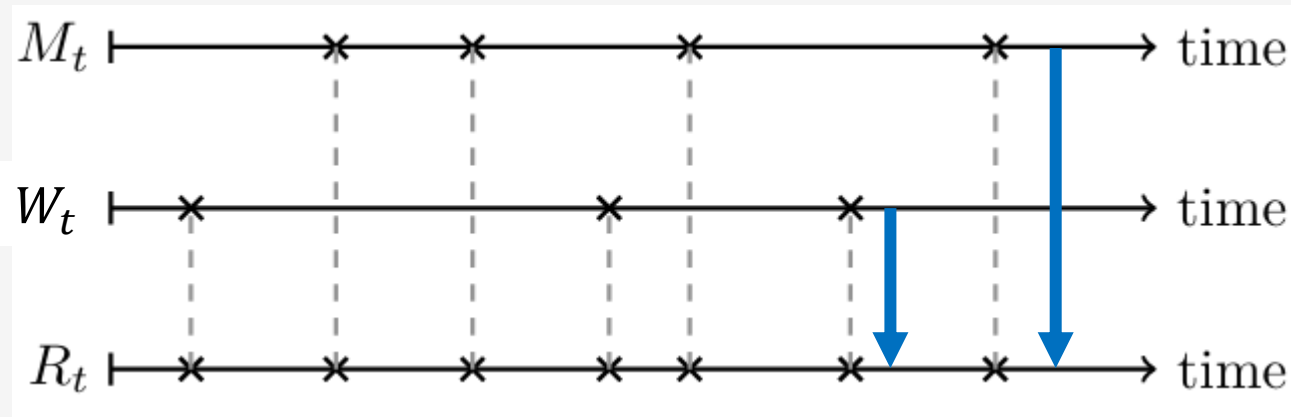
- Examples on Poisson Processes
 - Exponential RVs
 - Geometric sum of exponentials
 - Waiting for a bus
 - Kidney transplants
 - $\frac{1}{2}$ today, $\frac{1}{2}$ Wednesday

PSET 5: due Wednesday March 18, 23:59

Midterm 1 has been graded (almost) and grades will be released shortly.

Please reach out if you want to discuss anything!

Recap



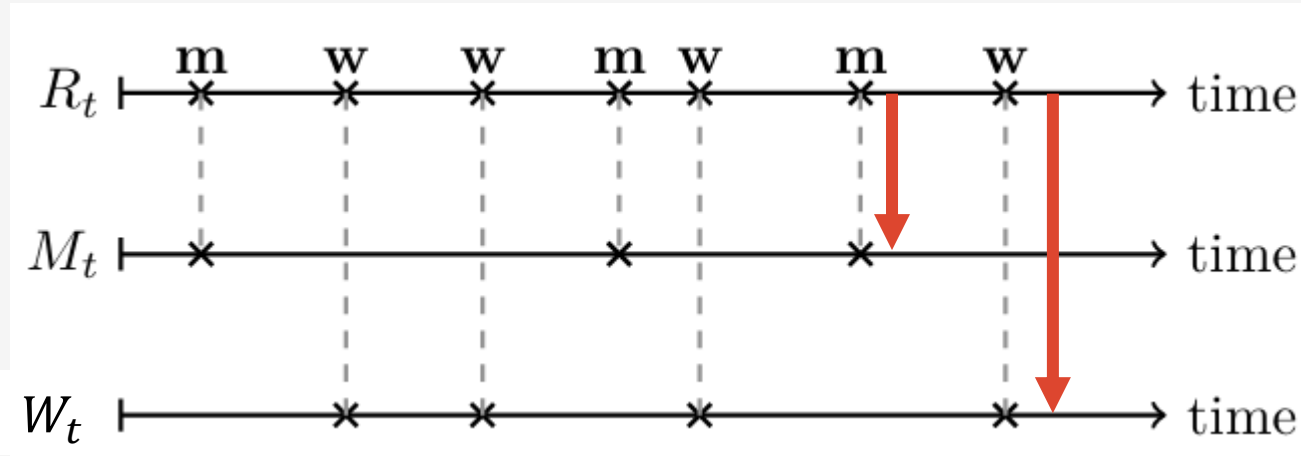
Principle of Superposition (or merging):

If we have k Poisson random processes $\{R_t^{(1)}\}_{t \geq 0}$, $\{R_t^{(2)}\}_{t \geq 0}$, ..., $\{R_t^{(k)}\}_{t \geq 0}$ with rates

$$\mu_1, \mu_2, \dots, \mu_k \quad \text{and} \quad R_t = R_t^{(1)} + R_t^{(2)} + \dots + R_t^{(k)}$$

Then $\{R_t\}_{t \geq 0}$ is a Poisson random process with **rate** $\mu = \mu_1 + \mu_2 + \dots + \mu_k$

Recap



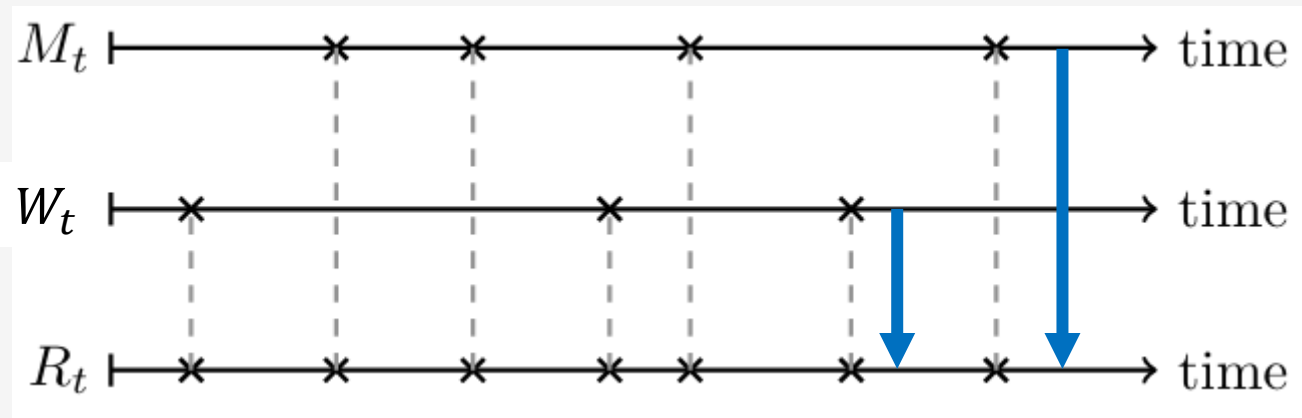
Principle of Thinning

If $\{R_t\}_{t \geq 0}$ is a Poisson process with rate μ ,

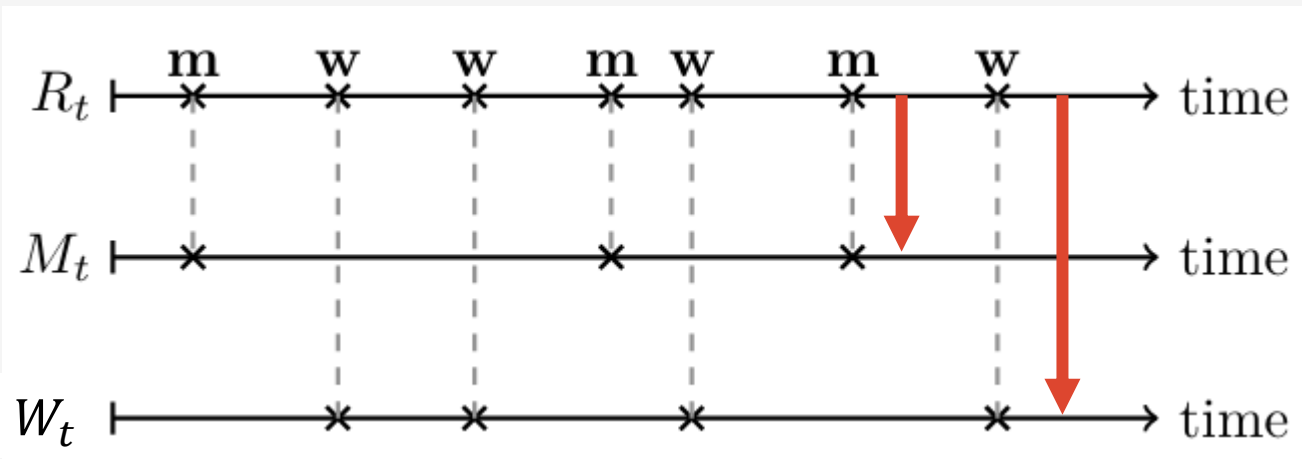
- and each arrival is assigned a type in the set $\{1, 2, \dots, k\}$ with probabilities $p_1, p_2, \dots, p_k$ such that $p_1 + p_2 + \dots + p_k = 1$,
- and $N_t^{(i)}$ = "# arrivals of type i by time t " for $i = 1, 2, \dots, k$,

Then $\{N_t^{(i)}\}_{t \geq 0}$ for $i = 1, 2, \dots, k$ are **independent** Poisson processes with **rates** $p_i \mu$

Recap



Superposition



Thinning

Examples

Exponential RVs

Exponential RVs

- **Example 1:**

- Consider $X \sim \text{Expon}(\mu)$ and $Y \sim \text{Expon}(\nu)$
- Recall that we have proved:

1. $X \wedge Y \sim \text{Expon}(\mu + \nu)$ ($X \wedge Y := \min\{X, Y\}$)

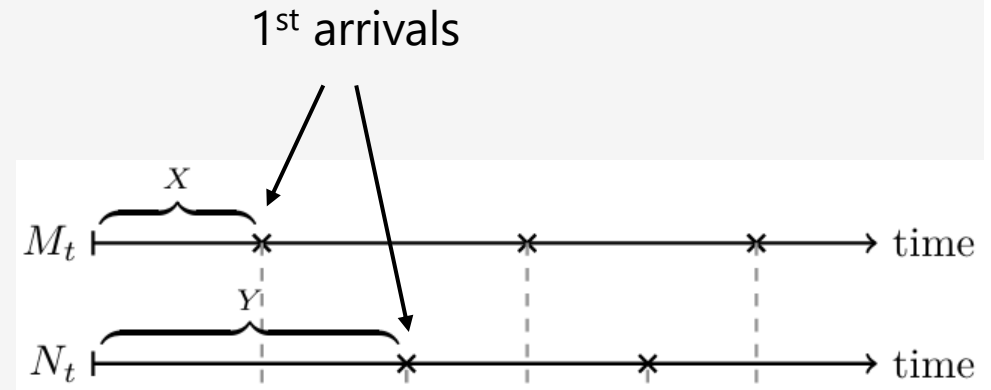
2. $P\{X \leq Y\} = \frac{\mu}{\mu + \nu}$

- This time we will use **Poisson Processes** to “**re-prove**” these results using different reasoning and take advantage of **Superposition** and **Thinning**

Exponential RVs

- **Example 1 (cont.):**

- Consider two independent Poisson processes
 - $\{M_t\}_{t \geq 0}$ with rate μ and $\{N_t\}_{t \geq 0}$ with rate ν
- Define two RVs



- X is the 1st arrival **time** of the Poisson process $\{M_t\}_{t \geq 0} \rightarrow X \sim \text{Expon}(\mu)$
- Y is the 1st arrival **time** of the Poisson process $\{N_t\}_{t \geq 0} \rightarrow Y \sim \text{Expon}(\nu)$

Exponential RVs

- **Example 1 (cont.):**
 - Consider also the **total arrivals** Poisson processes:
 - $\{R_t\}_{t \geq 0}$ with $R_t = M_t + N_t \sim \text{Pois}(\mu + \nu)$ (using **Superposition Principle**)

How can we think of $X \wedge Y := \min\{X, Y\}$ with respect to $\{R_t\}_{t \geq 0}$??

Exponential RVs

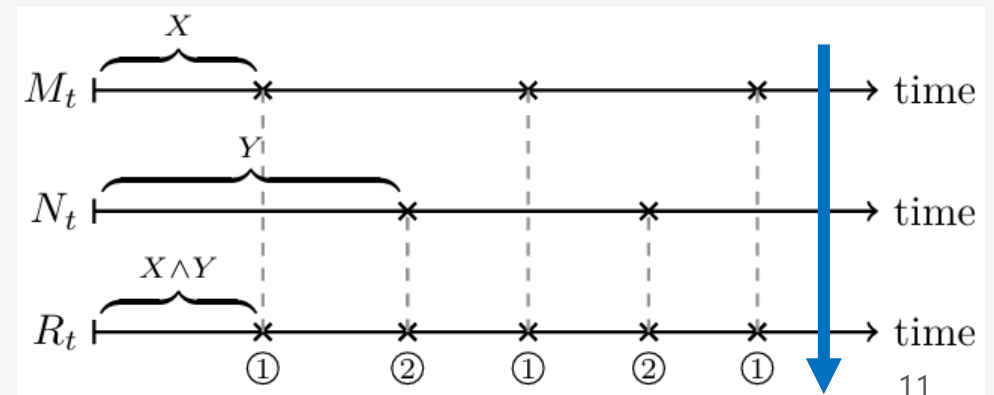
- **Example 1 (cont.):**

- Consider also the **total arrivals** Poisson processes:

- $\{R_t\}_{t \geq 0}$ with $R_t = M_t + N_t \sim \text{Pois}(\mu + \nu)$ (using **Superposition Principle**)

- We can think of $X \wedge Y$ as the **1st arrival time of the total arrivals** Poisson process $\{R_t\}_{t \geq 0}$

- Then $X \wedge Y \sim \text{Expon}(\mu + \nu)$



- **Example 1(cont.):**

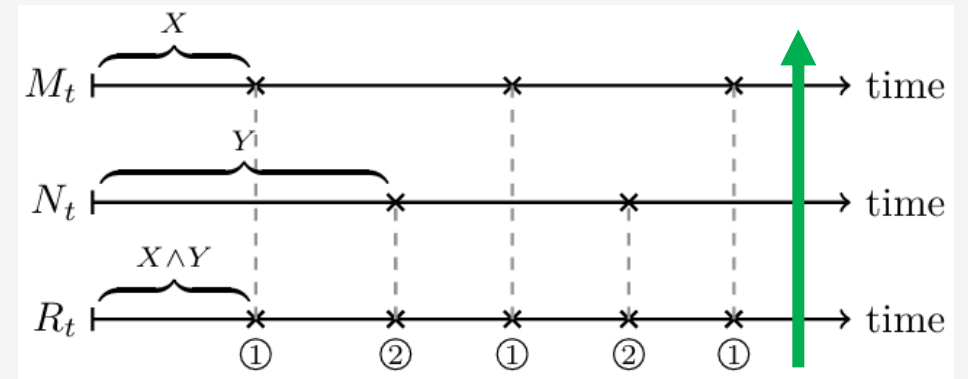
- Now let's compute $P\{X \leq Y\} \rightarrow$ We will use the **Thinning Principle** on $\{R_t\}_{t \geq 0}$
 - We **start** with the Poisson process $\{R_t\}_{t \geq 0}$ with rate $\mu + \nu$

- **Example 1(cont.):**

- Now let's compute $P\{X \leq Y\} \rightarrow$ We will use the **Thinning Principle** on $\{R_t\}_{t \geq 0}$

- We **start** with the Poisson process $\{R_t\}_{t \geq 0}$ with rate $\mu + \nu$

- Arrivals from $\{M_t\}_{t \geq 0}$ correspond to **(1)s** (a label)
- Arrivals from $\{N_t\}_{t \geq 0}$ correspond to **(2)s** (another label)



- **Example 1(cont.):**

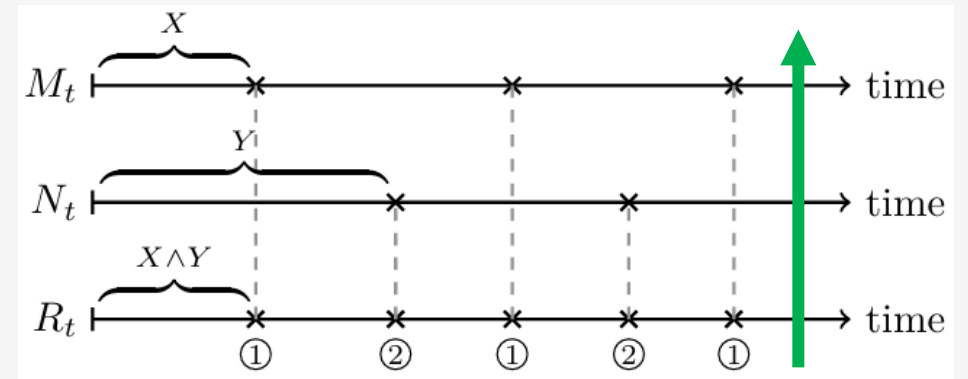
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- Define X_k as the label of the **k -th arrival** of $\{R_t\}_{t \geq 0}$



- **Example 1(cont.):**

- Now let's compute $P\{X \leq Y\} \rightarrow$ We will use the **Thinning Principle** on $\{R_t\}_{t \geq 0}$

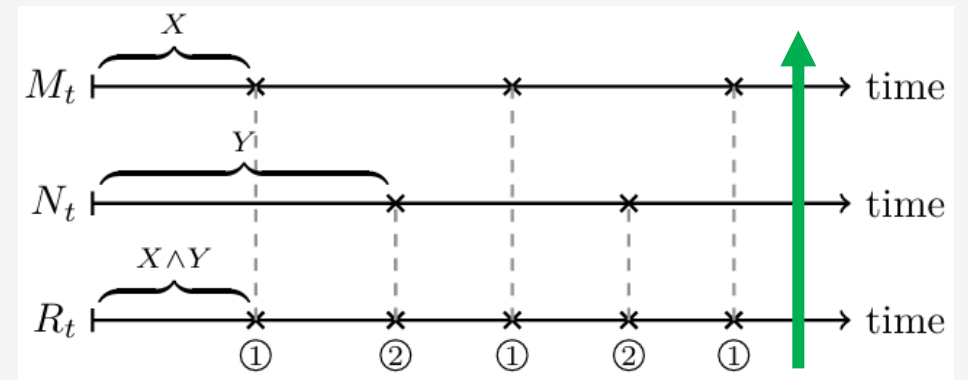
- We **start** with the Poisson process $\{R_t\}_{t \geq 0}$ with rate $\mu + \nu$

- Arrivals from $\{M_t\}_{t \geq 0}$ correspond to **(1)s** (a label)

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- Define X_k as the label of the **k -th arrival** of $\{R_t\}_{t \geq 0}$

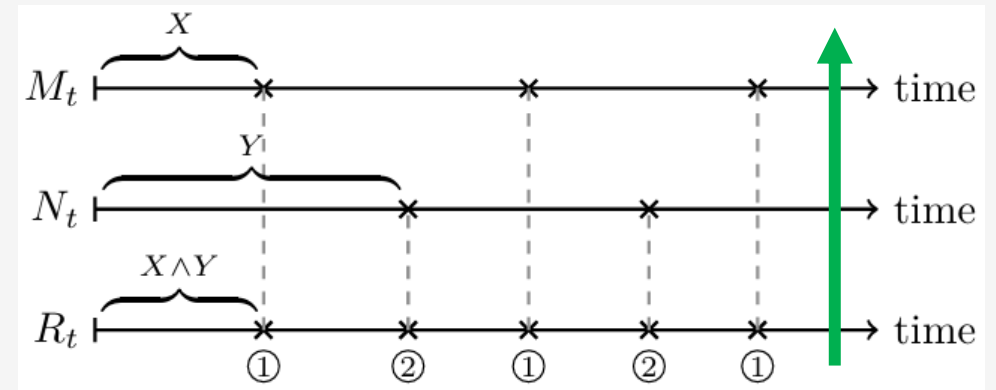
- Then $P\{X_k = (1)\} = \frac{\mu}{\mu + \nu}$ and $P\{X_k = (2)\} = \frac{\nu}{\mu + \nu} \rightarrow P\{X_k = (1)\} + P\{X_k = (2)\} = 1$



- **Example 1 (cont.):**

- Using the **Thinning principle**, we can derive that

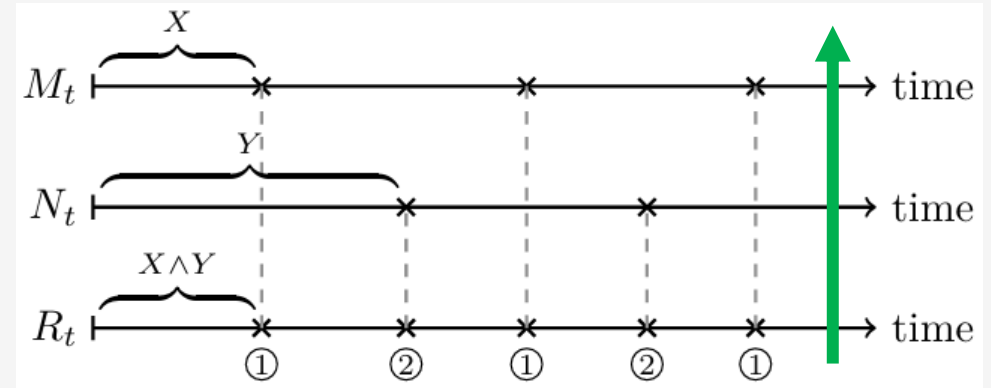
- $\{M_t\}_{t \geq 0}$: Poisson process with rate μ
- $\{N_t\}_{t \geq 0}$: Poisson process with rate ν



- **Example 1 (cont.):**

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- $\{M_t\}_{t \geq 0}$: Poisson process with rate μ
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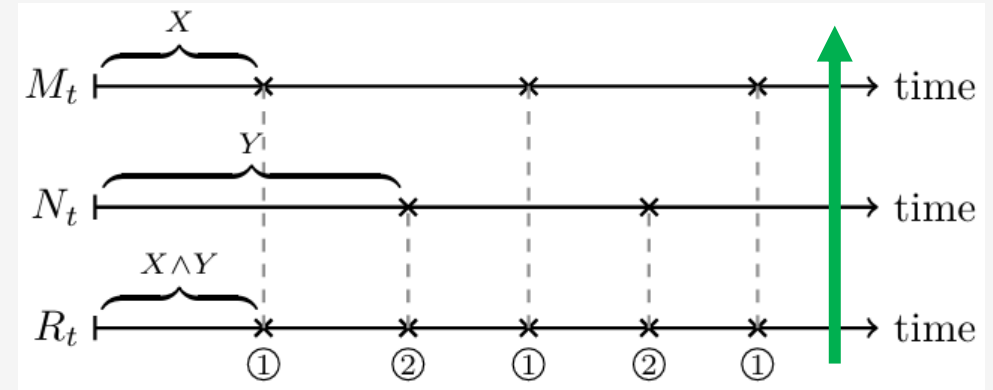


How can I write the event $\{X \leq Y\}$ as another event that involves $X \wedge Y$ and X ??

- **Example 1 (cont.):**

- Using the **Thinning principle**, we can derive that

- $\{M_t\}_{t \geq 0}$: Poisson process with rate μ
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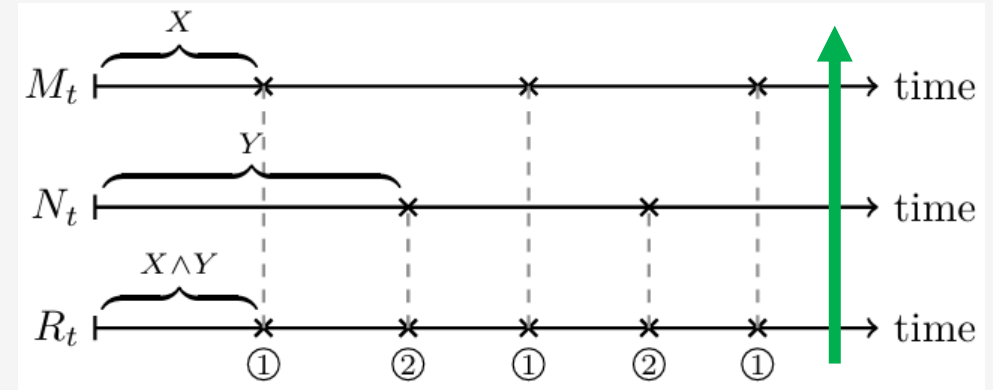


- The event $\{X \leq Y\}$ is **equivalent to** the event $\{X \wedge Y = X\}$
 - **If** $X \wedge Y := \min\{X, Y\} = X$ **then** we logically have $X \leq Y$

- **Example 1 (cont.):**

- Using the **Thinning principle**, we can derive that

- $\{M_t\}_{t \geq 0}$: Poisson process with rate μ
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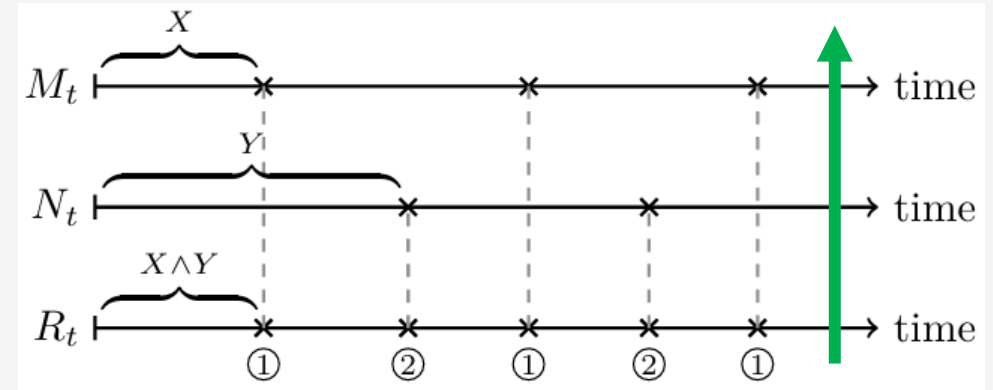


- The event $\{X \leq Y\}$ **is equivalent to** the event $\{X \wedge Y = X\}$
 - **If** $X \wedge Y := \min\{X, Y\} = X$ **then** we logically have $X \leq Y$
- Then $P\{X \leq Y\} = P\{X \wedge Y = X\} = P\{\text{first label is } (1)\} = P\{X_1 = (1)\} =$

- **Example 1 (cont.):**

- Using the **Thinning principle**, we can derive that

- $\{M_t\}_{t \geq 0}$: Poisson process with rate μ
- $\{N_t\}_{t \geq 0}$: Poisson process with rate ν



- The event $\{X \leq Y\}$ is **equivalent to** the event $\{X \wedge Y = X\}$

- **If** $X \wedge Y := \min\{X, Y\} = X$ **then** we logically have $X \leq Y$

- Then $P\{X \leq Y\} = P\{X \wedge Y = X\} = P\{\text{first label is } (1)\} = P\{X_1 = (1)\} = \frac{\mu}{\mu + \nu}$

Real life scenario 1: Reliability Engineering

- **Scenario:** a system S is functioning normally, if two independent devices, A and B function properly.
- Let $X \sim \text{Expon}(\mu)$ be the time till device **A fails**
- Let $Y \sim \text{Expon}(\nu)$ be the time till device **B fails**
- **Question:** What is the probability that device **A fails first**?
- **Answer:** $P\{X \leq Y\} = \mu/(\mu + \nu)$
- **Importance:** Identify the weakest link in a system, prioritize maintenance based on the highest probability of failure.

Real life scenario 2: Algorithmic trading – Order Book Dominance

- **Scenario:** a quantitative trading firm is running a dual-strategy algorithm on a single stock name, ready to execute a trade.
- Let $X \sim \text{Expon}(\mu)$ be the time till a **large Buy Order arrives** (pushing the stock price high)
- Let $Y \sim \text{Expon}(\nu)$ be the time till a **large Sell Order arrives** (pushing the stock price low)
- The trading algorithm executes its trades only when the next major-moving event occurs (large Buy/Sell Order)
- **Question:** What is the probability that the next market-moving event will be a Buy Order (so the algo buys more)
- **Answer:** $P\{X \leq Y\} = \mu / (\mu + \nu)$
- **Importance:** the company earns higher profit by only executing trades when the probability of winning money is sufficiently high

Geometric sum of exponentials

Geometric sum of exponentials

- **Example 2:**

- Consider the **i.i.d.** RVs: $Y_1, Y_2, Y_3, \dots$ with $Y_i \sim \text{Expon}(\mu)$, $i = 1, 2, \dots$
- Consider the RV: $N \sim \text{Geom}(p)$
- **Question:** what is the distribution of the new RV:

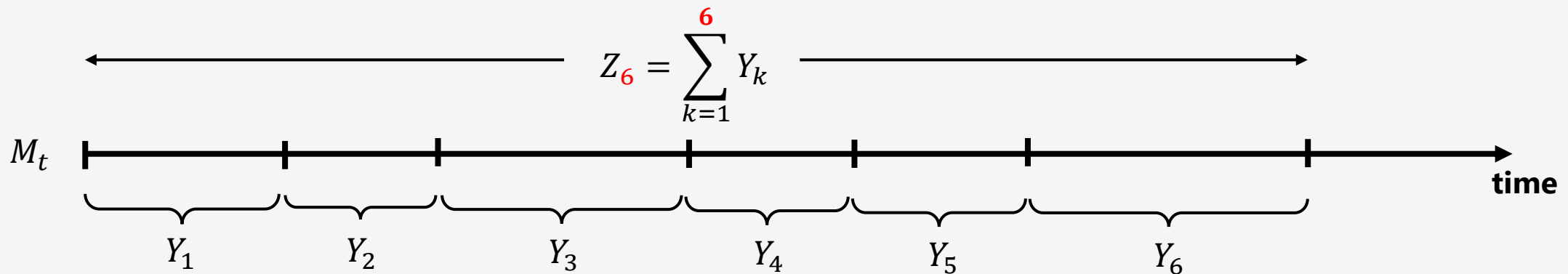
$$Z = \sum_{k=1}^N Y_k$$

We have to “engineer” the RVs within a Poisson process!

Geometric sum of exponentials

- **Example 2 (cont.):**
 - **Recall** the *inter-arrival times* in a Poisson process are **i.i.d. Exponential RVs**
 - We will define a Poisson process $\{M_t\}_{t \geq 0}$ using the RVs: $Y_1, Y_2, Y_3, \dots$

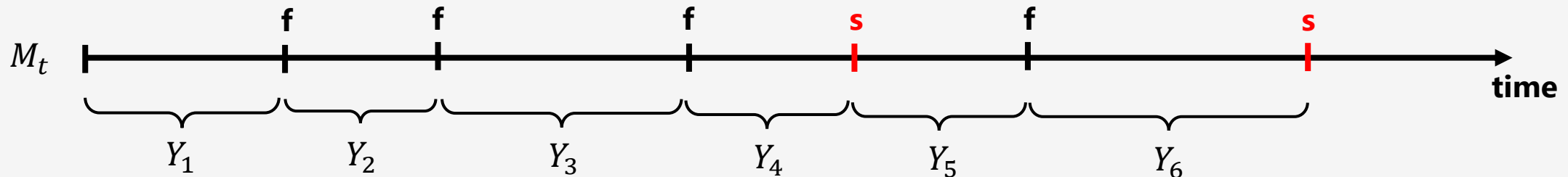
A realization of the RV Z when $N = 6$



Geometric sum of exponentials

- **Example 2 (cont.):**

- Recall that $N \sim \text{Geom}(p) \rightarrow$ So it is the trial or "arrival" the 1st of success
- We can attach labels s and f on each arrival in $\{M_t\}_{t \geq 0}$
 - With $P\{Y_k = s\} = p$ and $P\{Y_k = f\} = 1 - p$

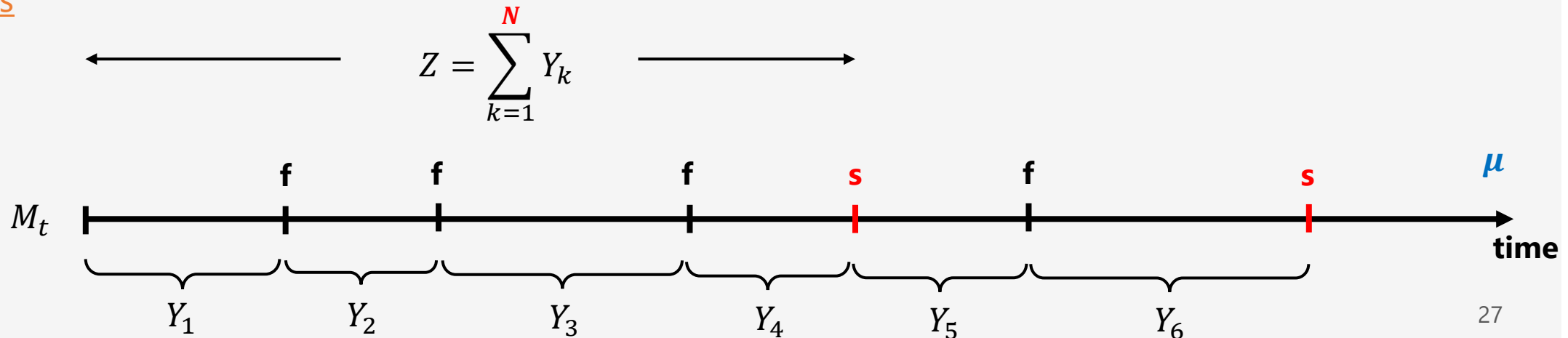


Geometric sum of exponentials

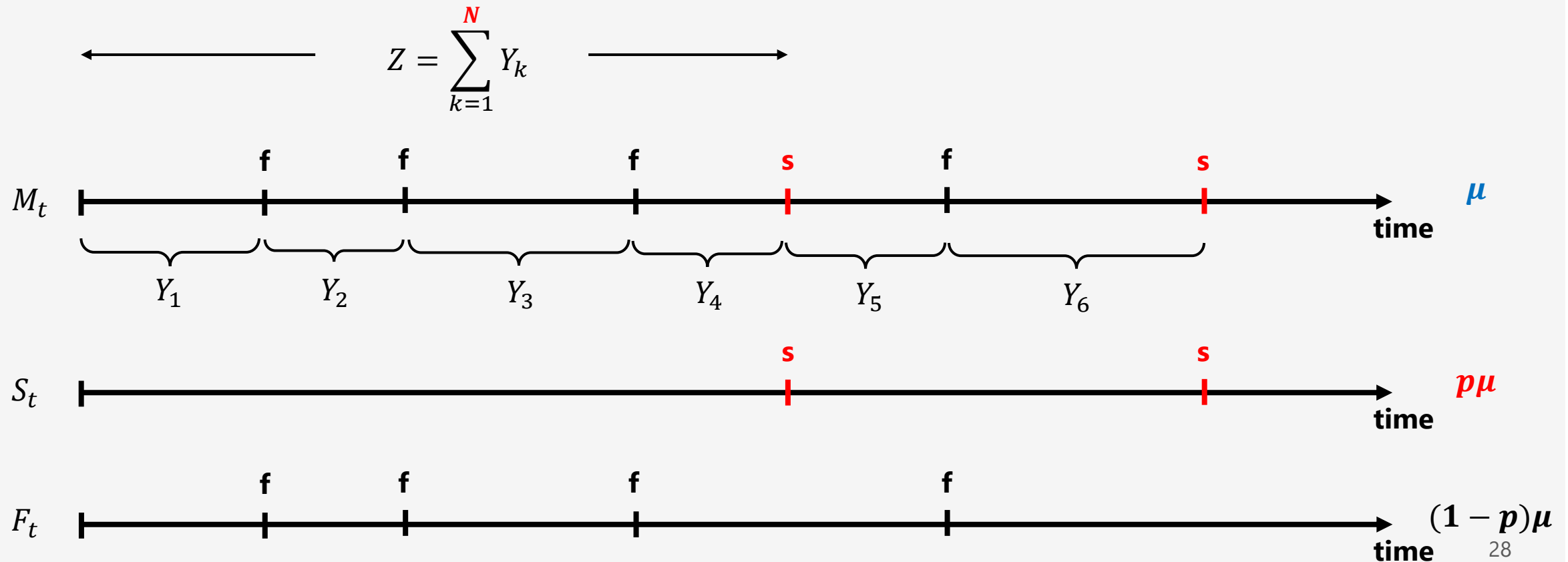
- **Example 2 (cont.):**

- So, the **original question**: what is the distribution of the RV: $Z = \sum_{k=1}^N Y_k$

can be **thought of** as follows: **If** we take a Poisson process $\{M_t\}_{t \geq 0}$ with rate μ , and at each arrival we associate a **success** with probability p and **failure** with probability $1 - p$, **then** what is the time of the arrival of the 1st success? This can be also **interpreted as**: the sum of the inter-arrival times until the 1st success



- **Example 2 (cont.):** the Thinning Principle for the Poisson process $\{M_t\}_{t \geq 0}$



Geometric sum of exponentials

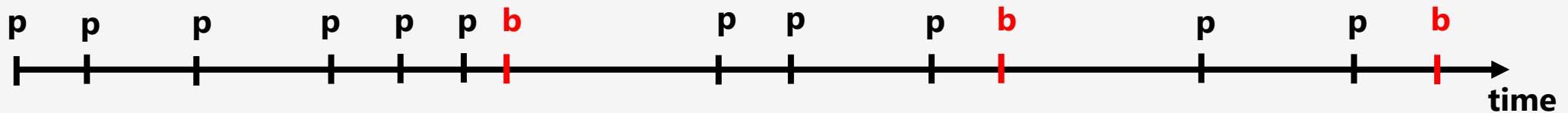
- **Example 2 (cont.):**
 - Now according the **Thinning Principle**, the **arrival of successes** $\{S_t\}_{t \geq 0}$ is a **Poisson process** with rate $p\mu$
 - And hence the **time of the arrival of the 1st success** follows an Exponential distribution with rate μp

$$Z = \sum_{k=1}^N Y_k \sim \text{Expon}(\mu p)$$

Waiting for a bus

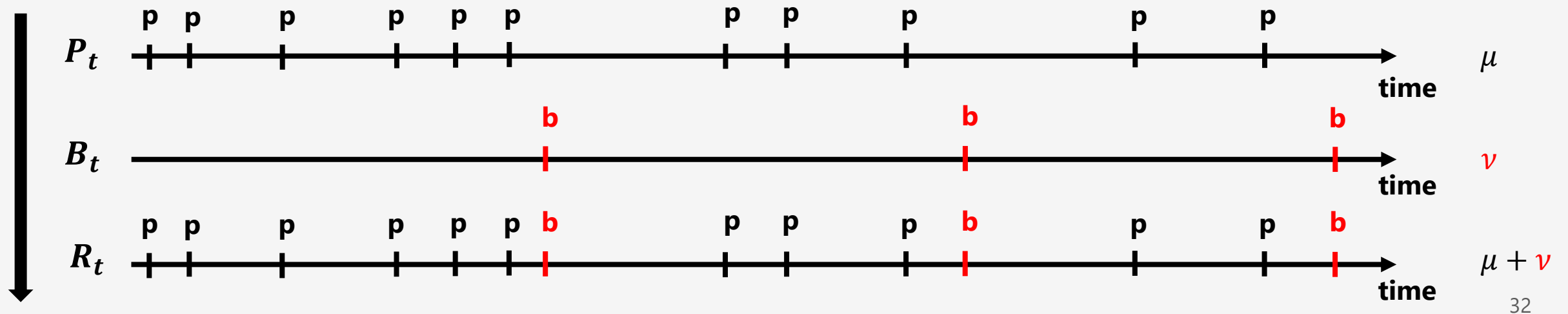
Waiting for a bus

- **Example 3:** *(slightly different from the notes)*
 - Passengers arrive at a bus stop according to a Poisson process with rate μ
 - Buses arrive at that bus stop according to a Poisson process with rate ν
 - **Question:** what is the distribution of the passengers that get on the 1st bus?



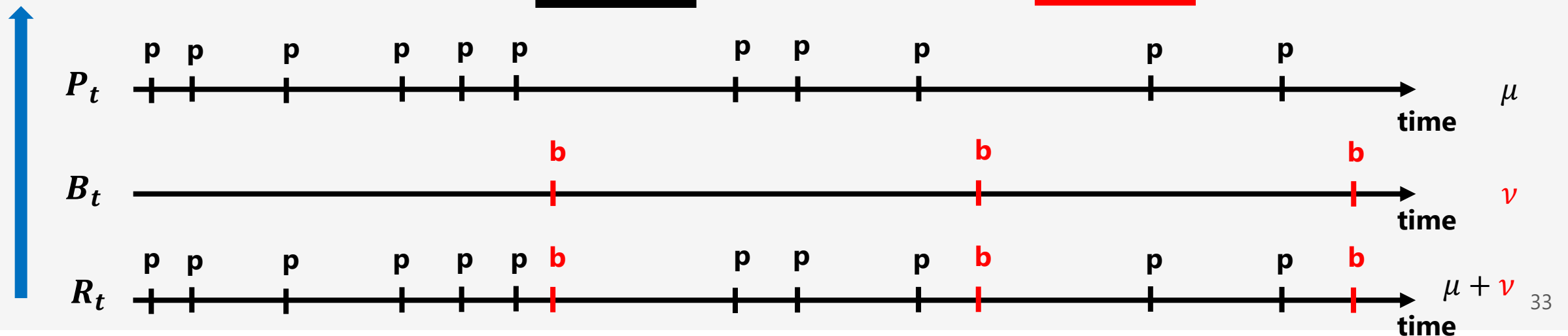
- **Example 3 (cont.):**

- We can define the Poisson processes for the passengers $\{P_t\}_{t \geq 0}$ and the buses $\{B_t\}_{t \geq 0}$ and using **Superposition** we can derive the Poisson process $\{R_t\}_{t \geq 0}$ of the passengers and busses



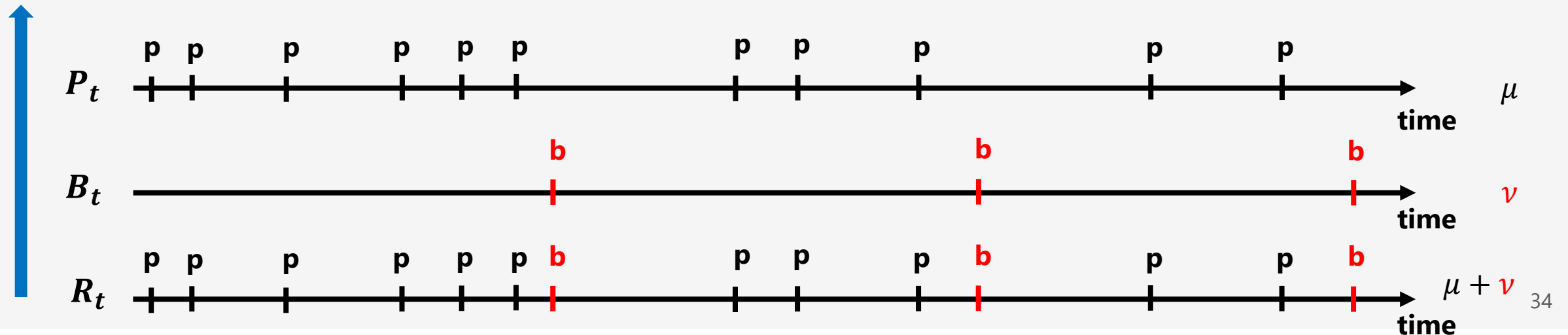
- **Example 3 (cont.):**

- We can now use **Thinning Principle** and assign probabilities to passengers and buses
- Define the RV: $X_k = \text{"the type of the } k\text{-th arrival"}$
- $P\{X_k = \text{Passenger}\} = \blacksquare$ and $P\{X_k = \text{Bus}\} = \blacksquare$



- **Example 3 (cont.):**

- We can now use **Thinning Principle** and assign probabilities to passengers and buses
- Define the RV: X_k = "the type of the k -th arrival"
- $P\{X_k = \text{Passenger}\} = \mu/(\mu + \nu)$ and $P\{X_k = \text{Bus}\} = \nu/(\mu + \nu)$



- **Example 3 (cont.):**

- Then we can consider **Passengers as Failures** and **Buses as Successes**

$$P\{k \text{ Passengers get on 1st Bus}\} =$$

- **Example 3 (cont.):**

- Then we can consider *Passengers as Failures* and *Buses as Successes*

$$\begin{aligned} P\{k \text{ Passengers get on 1st Bus}\} &= \\ &= P\{\textit{first } k \textit{ arrivals are Passengers}, (k + 1) \textit{ arrival is Bus}\} = \end{aligned}$$

- **Example 3 (cont.):**

- Then we can consider **Passengers as Failures** and **Buses as Successes**

$$\begin{aligned} P\{k \text{ Passengers get on 1st Bus}\} &= \\ &= P\{\text{first } k \text{ arrivals are Passengers, } (k+1) \text{ arrival is Bus}\} = \\ &= \left(\frac{\mu}{\mu + \nu}\right)^k \frac{\nu}{\mu + \nu} \end{aligned}$$

- Hence the number of Passengers that get on the 1st Bus $\sim \text{Geom}\left(\frac{\nu}{\mu + \nu}\right) - 1$

Remark: **we must subtract 1** from the standard Geometric distribution to account of the fact that there may be **no passenger** at the bus stop when the bus arrived.

Kidney transplants

Kidney transplants

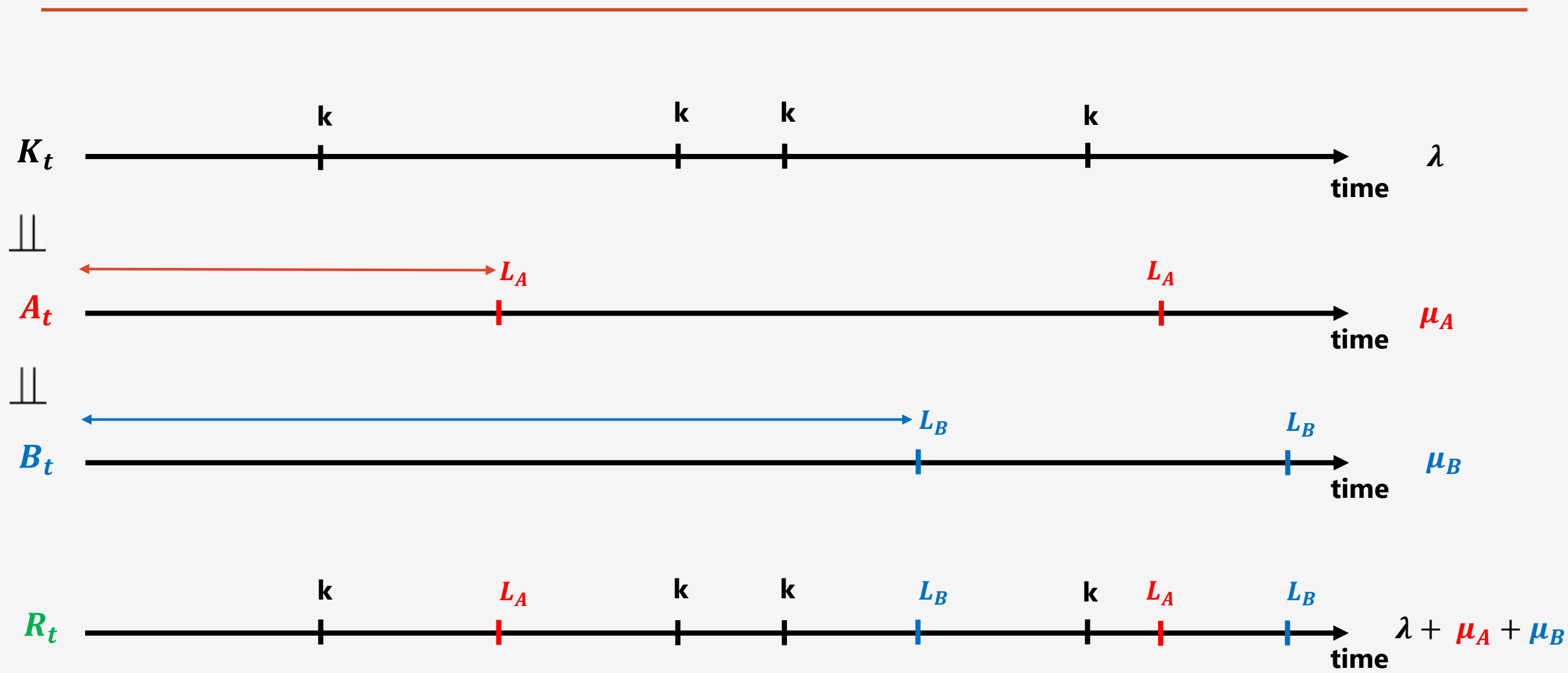
- **Example 4:**

- Each of the two patients A and B need to have a kidney transplant
- The **Lifetime of A** without kidney transplant is $L_A \sim \text{Expon}(\mu_A)$
- The **Lifetime of B** without kidney transplant is $L_B \sim \text{Expon}(\mu_B)$
- L_A and L_B are **independent**
- Kidneys arrive according to a **Poisson process** with rate λ
- **Policy:** Patient A has priority (A has shorter expected time to survive than B ($\mu_A > \mu_B$))

-
- **Example 4 (cont.):**
 - **Question:** what is the probability that B takes the kidney before he dies.
 - **Easy to think:** kidneys arrivals follow a Poisson process with rate λ
 - **Requires more thinking:** the lifetimes of patients A and B cannot be modeled as Poisson process, because everyone dies once

- **Example 4 (cont.):**

- **Question:** what is the probability that B takes the kidney before he dies.
- **Easy to think:** kidneys arrivals follow a Poisson process with rate λ
- **Requires more thinking:** the lifetimes of patients A and B cannot be modeled as Poisson process, because everyone dies once
- **BUT** we can think of Lifetime L_A as the **first arrival of death** for patient A ; and the Lifetime L_B of B as the **first arrival of death** for patient B
 - Then $\{A_t\}_{t \geq 0}$ and $\{B_t\}_{t \geq 0}$ are the Poisson processes of the **deaths** of A and B



$R_t = K_t + A_t + B_t$ (Superposition gives the total arrivals)

-
- Using **Thinning** on the Poisson process $\{R_t\}_{t \geq 0}$ we can define each process $\{K_t\}_{t \geq 0}$, $\{A_t\}_{t \geq 0}$ and $\{B_t\}_{t \geq 0}$ by randomly labeling the arrivals in $\{R_t\}_{t \geq 0}$ and assigning the following **probabilities**
 - $P\{\text{"kidney"}\} = \lambda / (\lambda + \mu_A + \mu_B)$
 - $P\{\text{"death of A"}\} = \mu_A / (\lambda + \mu_A + \mu_B)$
 - $P\{\text{"death of B"}\} = \mu_B / (\lambda + \mu_A + \mu_B)$

-
- **Remember:** our question is to find the probability of the **event**:

$$E = \{B \text{ gets kidney before he dies}\}$$

- **Remember:** patient A takes priority over patient B
- To find $P\{E\}$ we have to think of **all possible cases**
- **Remember:** a person is removed from the list if a **kidney** arrives or if he/she **dies**
- We have 2 patients, so we are looking for two events...

-
1. **Patient A is removed from the list if** the first arrival in $\{\mathbf{R}_t\}_{t \geq 0}$ is either $\mathbf{k}$ (a kidney has arrived) or $\mathbf{L}_A$ (patient A has died)
 2. When Patient A is removed from the list, **Patient B will live if a kidney arrives before $\mathbf{L}_B$ (i.e., before he dies)**
- Hence, we have identified the **two events** that make up the original event $\mathbf{E}$

$$\left. \begin{aligned} &\bullet \mathbf{E}_1 = \{\text{the 1st arrival in } R_t \text{ is } \mathbf{k} \text{ or } \mathbf{L}_A\} \\ &\bullet \mathbf{E}_2 = \{\text{in all next arrivals, } \mathbf{k} \text{ arrives before } \mathbf{L}_B\} \end{aligned} \right\} \begin{aligned} &\Rightarrow \mathbf{E} = \mathbf{E}_1 \cap \mathbf{E}_2 \\ &\Rightarrow P\{\mathbf{E}\} = P\{\mathbf{E}_1 \cap \mathbf{E}_2\} \end{aligned}$$

-
- **CLAIM:** events E_1 and E_2 are independent.
 - Thinning principle $\rightarrow$ the labels of the arrivals are independent.
 - E_1 depends on the labels of the **1st arrival only**
 - E_2 depends on the labels of the **2nd , 3rd ,... arrivals only**
 - **Hence** E_1 and E_2 are independent
 - We can therefore compute the probability: $P\{E\} = P\{E_1 \cap E_2\} = P\{E_1\}P\{E_2\}$

-
- Let's compute the individual probabilities $P\{E_1\}$ and $P\{E_2\}$:
 - $P\{E_1\}$ is straightforward to compute

$$P\{E_1\} = P\{\text{the 1}^{\text{st}} \text{ arrival in } \mathbf{R_t} \text{ is } \mathbf{k} \text{ or } \mathbf{L_A}\} =$$

$$= P\{R_t = "k"\} + P\{R_t = L_A\} = \frac{\lambda}{\lambda + \mu_A + \mu_B} + \frac{\mu_A}{\lambda + \mu_A + \mu_B}$$

- **$P\{E_2\}$ requires more thinking...**

-
- **Let's do this thinking...**

$$P\{E_2\} = P\{\text{in all next arrivals, "k" arrives before } L_B\} =$$

-
- **Let's do this thinking...**

$$\begin{aligned} P\{E_2\} &= P\{\text{in all next arrivals, "k" arrives before } L_B\} = \\ &= P\{\text{in arrivals 2,3,..., a "k" arrives before an } L_B\} = \end{aligned}$$

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- **Let's do this thinking...**

$$\begin{aligned} P\{E_2\} &= P\{\text{in all next arrivals, "k" arrives before } L_B\} = \\ &= P\{\text{in arrivals 2,3,..., a "k" arrives before an } L_B\} = \\ &= P\{\text{in arrivals } \mathbf{1},2,3,\dots, \text{ a "k" arrives before an } L_B\} = \end{aligned}$$

-
- **Let's do this thinking...**

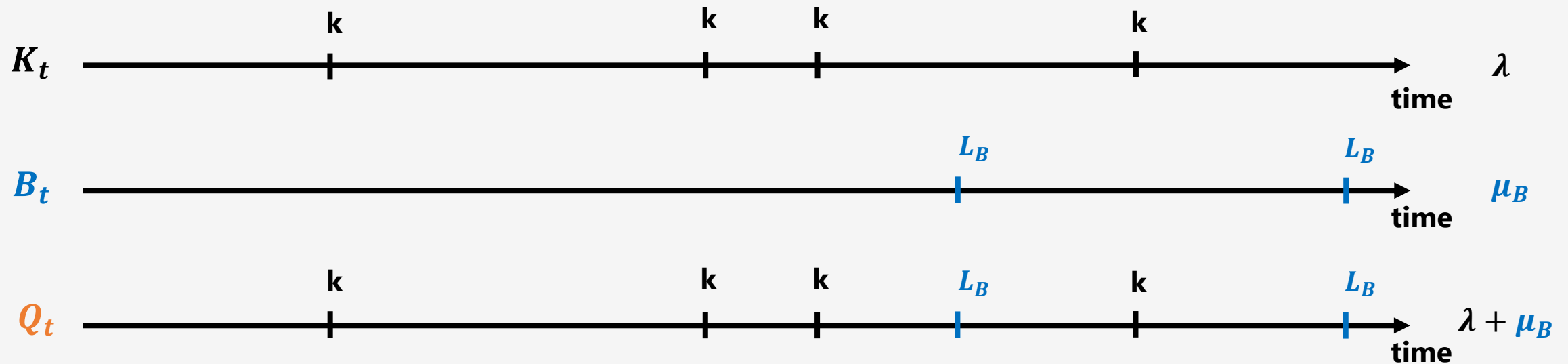
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- **Let's do this thinking...**

$$\begin{aligned} P\{E_2\} &= P\{\text{in all next arrivals, } k \text{ arrives before } L_B\} = \\ &= P\{\text{in arrivals } 2, 3, \dots, \text{ a "k" arrives before an } L_B\} = \\ &= P\{\text{in arrivals } \mathbf{1}, 2, 3, \dots, \text{ a "k" arrives before an } L_B\} = \\ &= P\{\text{a "k" arrives before an } L_B\} \end{aligned}$$

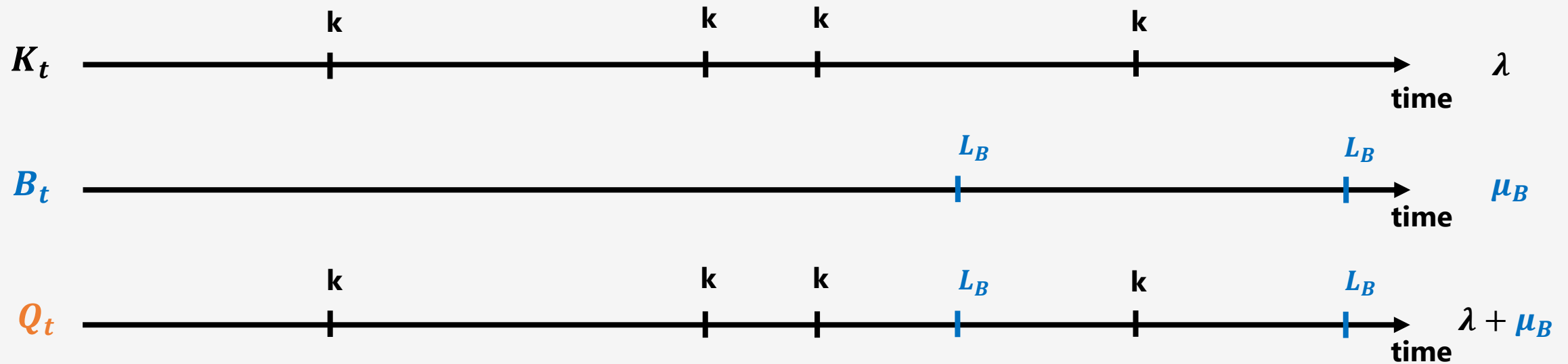
- So, we are **only** interested in the **labels "k" and L_B** in the Poisson process $\{R_t\}_{t \geq 0}$

- Using **Superposition** we define the **new** Poisson process $\{Q_t\}_{t \geq 0}$ where $Q_t = K_t + B_t$
- Then we apply the **Thinning Principle** on $\{Q_t\}_{t \geq 0}$



Question: What is the probability of a " k " being the 1st arrival in $\{Q_t\}_{t \geq 0}$??

- Using **Superposition** we define the Poisson process $\{Q_t\}_{t \geq 0}$ where $Q_t = K_t + B_t$
- Then we apply the **Thinning Principle** on $\{Q_t\}_{t \geq 0}$



Question: What is the probability of a " k " being the 1st arrival in $\{Q_t\}_{t \geq 0}$??

Answer: $P\{E_2\} = P\{\text{a "k" arrives before an } L_B\} = P\{\text{the 1st arrival in } Q_t \text{ is a "k"}\} = \lambda / (\lambda + \mu_B)$

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- Let's **summarize** what we have got:

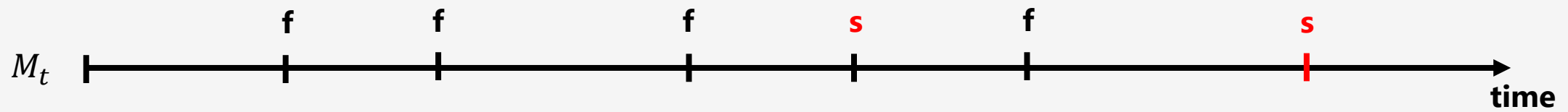
- $P\{E\} = P\{E_1 \cap E_2\} = P\{E_1\}P\{E_2\}$

- $P\{E_1\} = \frac{\lambda}{\lambda + \mu_A + \mu_B} + \frac{\mu_A}{\lambda + \mu_A + \mu_B} = \frac{\lambda + \mu_A}{\lambda + \mu_A + \mu_B}$

- $P\{E_2\} = \frac{\lambda}{\lambda + \mu_B}$

- Hence:** $P\{E\} = \frac{\lambda + \mu_A}{\lambda + \mu_A + \mu_B} \frac{\lambda}{\lambda + \mu_B}$ 

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